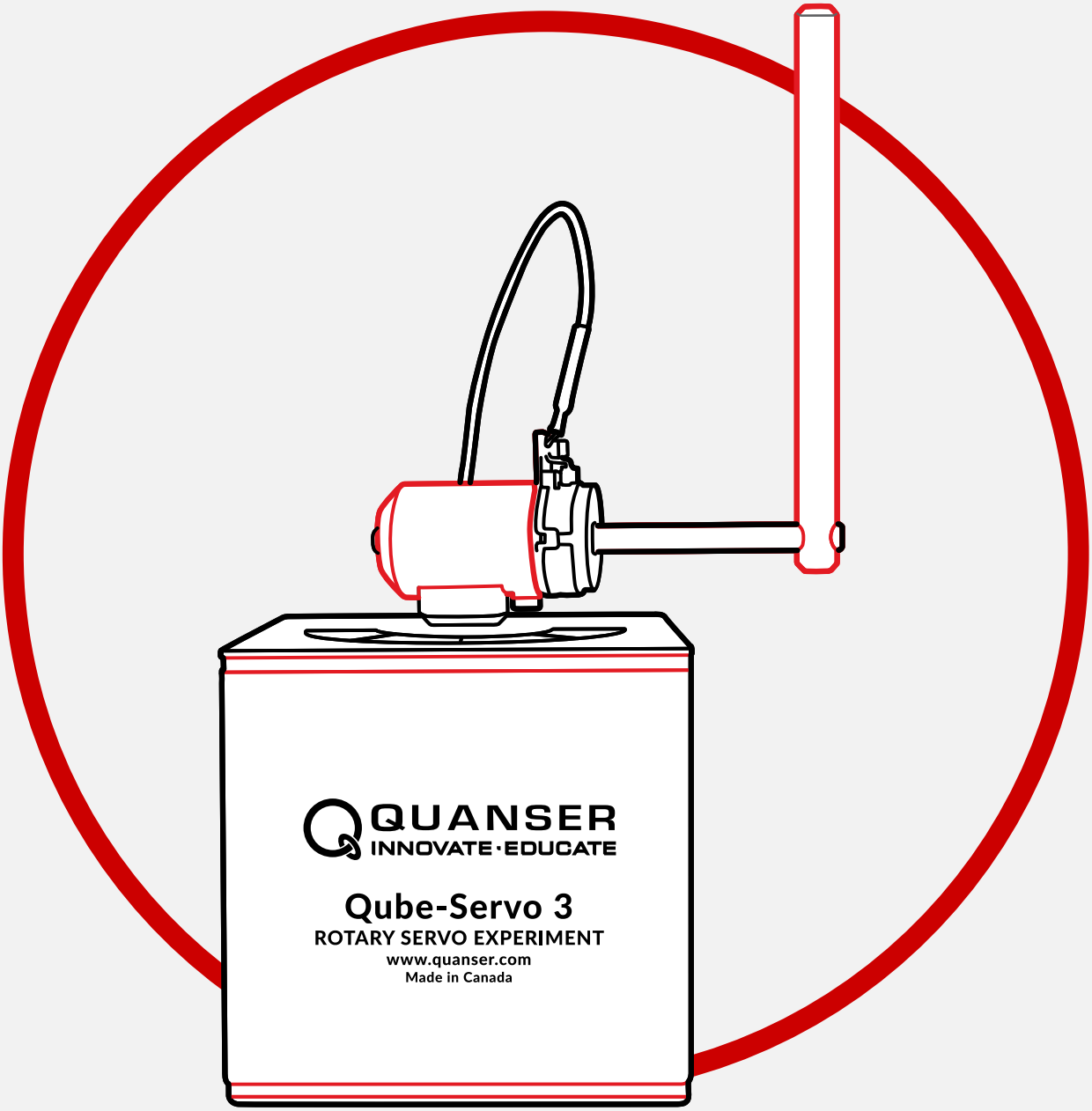


Qube-Servo 3

Optimal LQR Balance Control

V1.2 – 27th February 2025

© 2025 Quanser Consulting Inc., All rights reserved.
For more information on the solutions Quanser offers,
please visit the web site at: http://www.quanser.com

Quanser Consulting Inc. info@quanser.com
119 Spy Court Phone : 19059403575
Markham, Ontario Fax : 19059403576
L3R 5H6, Canada printed in Markham, Ontario.

This document and the software described in it are provided subject to a license agreement. Neither the software nor this document may be used or copied except as specified under the terms of that license agreement. Quanser Consulting Inc. ("Quanser") grants the following rights: a) The right to reproduce the work, to incorporate the work into one or more collections, and to reproduce the work as incorporated in the collections, b) to create and reproduce adaptations provided reasonable steps are taken to clearly identify the changes that were made to the original work, c) to distribute and publicly perform the work including as incorporated in collections, and d) to distribute and publicly perform adaptations. The above rights may be exercised in all media and formats whether now known or hereafter devised. These rights are granted subject to and limited by the following restrictions: a) You may not exercise any of the rights granted to You in above in any manner that is primarily intended for or directed toward commercial advantage or private monetary compensation, and b) You must keep intact all copyright notices for the Work and provide the name Quanser for attribution. These restrictions may not be waved without express prior written permission of Quanser.

Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.
Japan VCCI Notice This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI). If this equipment is used in a domestic environment, radio interference may occur, in which case the user may be required to take corrective actions.

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI-A

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

电子信息产品污染控制管理办法 (中国 RoHS)

中国客户Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

CE Compliance CE

This product meets the essential requirements of applicable European Directives as follows:

- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Qube-Servo 3 – Application Guide

Optimal LQR Balance Control

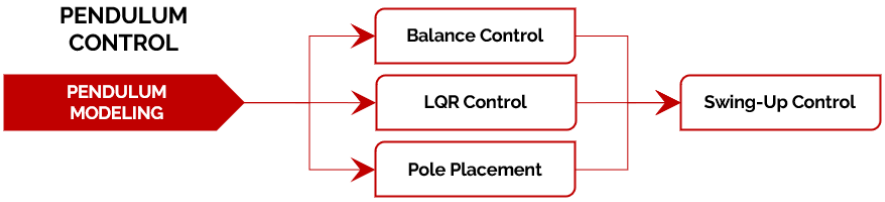
Why explore LQR Control?

Linear Quadratic Regulator (LQR) control is a powerful method for controlling systems with multiple inputs and outputs (MIMO) when the system can be described in state-space form and is operating in regions where linear approximations are valid. The approach finds an optimal feedback control law by minimizing a quadratic cost function that balances the tradeoff between state errors and control effort through user-defined weighting matrices. For pendulum systems and other applications, LQR effectively handles linearized dynamics around equilibrium points (including unstable ones) while allowing engineers to tune the system response characteristics through systematic adjustment of the cost function weights, provided the system is controllable.

Background

This lab is part of the Pendulum Control skills progression of the Qube-Servo 3. These labs are focused on understanding different ways to maintain balance of an inverted pendulum and finish off with creating an energy based controller to swing it up and then maintain the balance.

The lab progression is as follows:



Prior to starting this lab, please review the following concept reviews (should be located in Documents/Quanser/concept_reviews/),

- Concept Review – Modeling → Pendulum Free Body Diagram (Energy Control Section)

Linear Quadratic Regulator (LQR)

Linear Quadratic Regulator (LQR) theory is a technique that is ideally suited for finding the parameters of the pendulum balance controller.

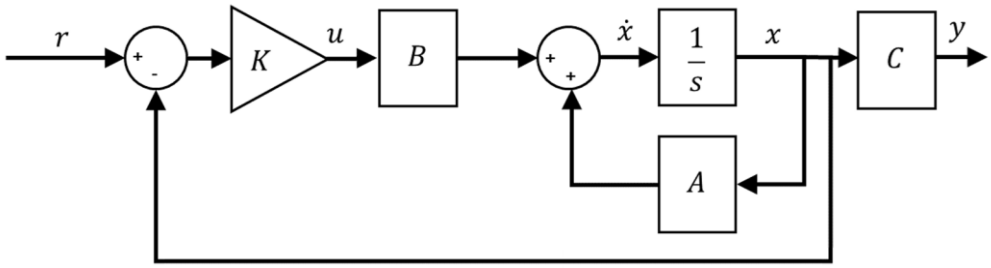
The standard state space representation of a multi input multioutput (MIMO) continuous linear time invariant (LTI) system with n state variables, r input variables, and m output variables is

$$\dot{x}(t) = Ax + Bu$$

$$y(t) = Cx(t) + Du(t)$$

Where:

- $x \in \Re^{n \times 1}$ is the vector of state variables
- $u \in \Re^{r \times 1}$ is the control input vector
- $y \in \Re^{m \times 1}$ is the output vector
- $A \in \Re^{n \times n}$ is the system matrix
- $B \in \Re^{n \times r}$ is the input matrix
- $C \in \Re^{m \times n}$ is the output matrix
- $D \in \Re^{m \times r}$ is the feed-forward matrix



The state feedback control of a system is shown in the figure above. If u is the state feedback control law, it has the form of

$$u = r - Kx$$

Without a reference signal: $u = -Kx$. Where $K \in \Re^{m \times n}$ is the feedback gain. Applying this to the state space equations gives the closed loop system

$$\dot{x}(t) = Ax + Bu = Ax - BKx = (A - BK)x$$

The state vector x of the rotary pendulum system is defined as

$$x = [\theta \quad \alpha \quad \dot{\theta} \quad \dot{\alpha}]^T$$

The reference signal is the desired rotary arm/base position and is defined as

$$x_{ref} = [\theta_r \quad 0 \quad 0 \quad 0]^T$$

The LQR algorithm computes a control law u such that the performance criterion or cost function below is minimized.

$$J = \int_0^\infty \left(x_{ref} - x(t)\right)^T Q \left(x_{ref} - x(t)\right) + u(t)^T R u(t) dt$$

The design matrices Q and R hold the penalties on the deviations of state variables from their setpoint and the control actions, respectively. When an element of Q is increased, therefore, the cost function increases the penalty associated with any deviations from the desired setpoint of that state variable, and thus the specific control gain will be larger. When the values of the R matrix are increased, a larger penalty is applied to the aggressiveness of the control action, and the control gains are uniformly decreased.

The control straregy used to balance the pendulum and track the rotary arm/base setpoint becomes

$$u = K\left(x_{ref} - x\right) = k_{p,\theta}(\theta_r - \theta) - k_{p,\alpha}\alpha - k_{d,\theta}\dot{\theta} - k_{d,\alpha}\dot{\alpha}$$

This control law is a state feedback control and is illustrated in the figure above. The structure is equivalent to the PD control used to balance the pendulum in the Balance Control lab.

Getting started

In this lab you will implement LQR control on the Qube-Servo 3 using the MATLAB *Control System Toolbox*. Using this toolbox, you will gain an intuitive understanding of how modifying the weighting matrices in LQR control affects the control gains calculated. Using this knowledge, you will then deploy the LQR controller to the Qube-Servo 3 and tune the weighting matrices to achieve certain performance criteria when balancing the pendulum.

Ensure you have completed the following labs

- **SP5_ Pendulum Modeling Labs**
- **Balance Control**

Before you begin this lab, ensure that the following criteria are met.

- If using a physical Qube-Servo 3, make sure it has been setup and tested. See the Qube-Servo 3 Quick Start Guide for details on this step.
- Make sure the pendulum attachment is set up and connected to the Qube-Servo 3 using the cable to the Encoder 1 port. Turn the plug to make sure the pendulum is centered around the front of the Qube at 0˚. The resistance from the cable will help keep it in the desired position.
- If using the virtual Qube-Servo 3, make sure you have Quanser Interactive Labs open in the Qube 3 - Pendulum → Pendulum Workspace.
- You are familiar with the basics of Simulink. See the Simulink Onramp for more help with getting started with Simulink.